

Module 3: Research Ethics, IPR, and Scholarly Publishing

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Ethics in Research

Research ethics involve principles that ensure honesty, integrity, and respect for all participants in a study.

A. Ethical Issues in Research

1. Informed Consent

- **Participants must voluntarily agree** to participate after being fully informed about the research.
- Should include: purpose, procedures, risks, benefits, and right to withdraw.
- **Example:** Signing a consent form before a medical trial.

2. Confidentiality and Privacy

- Participants' **personal information must be protected**.
- Data should be **anonymized or encrypted**.
- Researchers must not disclose identifiable info without permission.

3. Plagiarism

- **Using someone else's work** (words, ideas, data) **without credit** is unethical.
- Proper **citation** is essential.
- Even **self-plagiarism** (reusing your own previous work) should be disclosed.

4. Fabrication and Falsification

- **Fabrication:** Making up data or results.
- **Falsification:** Manipulating research processes or data.
- Both are serious misconducts.

5. Harm to Participants

- Researchers must **avoid physical, psychological, emotional, or social harm**.
- Special care for vulnerable groups (children, disabled, elderly).

6. Deception

- Should be **avoided unless necessary**, and participants must be **debriefed** afterward.
- Acceptable only when no alternatives are available and justified by study's value.

7. Conflict of Interest

- Researchers must **disclose any personal or financial interest** that could bias the study.
- Transparency ensures trustworthiness.

8. Intellectual Property Rights

- **Proper attribution** to co-authors, collaborators, or contributors.
- Avoid misrepresentation of contributions.

9. Ethical Approval

- Many studies (especially involving humans/animals) require **clearance from an ethics committee** or Institutional Review Board (IRB).

B. Ethical Committees (Human & Animal Research)

- **Institutional Review Board (IRB) / Ethical Committee** ensures research is ethical.
- **For Human Research:**
 - Informed **consent** is mandatory.
 - Participants should not face harm.
 - **Privacy & confidentiality** must be protected.
- **For Animal Research:**
 - Ethical treatment of animals is required.
 - Alternatives to animal testing should be considered.
 - Must follow guidelines set by **CPCSEA (India)** or similar bodies.

Scholarly Publishing

Publishing research helps disseminate knowledge and contributes to scientific advancements.

1. Importance of Publishing in Research

- **Academic Recognition** – Enhances credibility and career growth.
- **Knowledge Sharing** – Helps in the advancement of science and technology.
- **Funding Opportunities** – Strong publications can attract research grants.
- **Citations & Impact** – Increases influence in the research community.

2. Peer-Reviewed Journals: Research papers are reviewed by experts before publication. **Process:**

- Researcher submits a manuscript.
- Experts review for accuracy and originality.
- Reviewers suggest revisions or reject the paper.
- If accepted, the paper is published.
- **Importance:** Ensures quality, reliability, and validity of research.

3. Types of Publications

- **Journal Articles:** Published in scientific journals, often peer-reviewed.
- **Conference Papers:** Presented at conferences and sometimes published in proceedings.
- **Theses/Dissertations:** Submitted as part of academic degree requirements.
- **Books & Book Chapters:** Comprehensive research compilations.
- **Review Papers:** Summarize existing research in a field.
- **Technical Reports:** Research reports funded by government or organizations.

Intellectual Property Rights (IPR)

Intellectual Property refers to creations of human mind that have commercial value, and hence could be sold, rent. eg, company logo, a patented mobile processor design, novel or film script

IPR is the legal rights granted to individuals/ organizations to protect IP from unauthorized use or duplication

1. Importance of IPR in Research: IPR protects innovations, inventions, and creative works by ensuring legal rights to the creator. It helps in:

- **Encouraging innovation** – Researchers are motivated to create new knowledge.
- **Preventing unauthorized use** – Protects research from plagiarism and misuse.
- **Promoting commercialization** – Helps in monetizing research discoveries.
- **Facilitating collaborations** – Encourages research partnerships with industry.

Types of IPR

- a) **Copyright:** Legal right given to creators for their **original works** such as literary, artistic, musical, dramatic, software, photography, etc. It is the right to “not copy”
- **Purpose:** To **protect creative expression** and prevent unauthorized copying or distribution.
 - **Validity:** Typically, + 60 years (varies by country).
 - **Example:** A researcher’s published paper is protected by copyright.
 - **Key Rights Granted:**
 - Right to reproduce (copy) the work
 - Right to create derivative works
 - Right to distribute copies
 - Right to publicly display or perform the work
- b) **Patent:** A legal right granted to an inventor for a **new, non-obvious, and industrially applicable invention** (product or process).
- **Duration:** Grants exclusive rights for 20 years
 - **Purpose:** To promote technological innovation by protecting inventors from imitation.
 - **Example:** A new drug formulation patented by a scientist.
 - **Patentability Criteria:**
 - **Novelty** – Must be new
 - **Inventive Step** – Non-obvious to experts in the field
 - **Industrial Applicability** – Usable in industry
 - **What Cannot Be Patented:**
 - Scientific theories
 - Agricultural methods
 - Traditional knowledge
 - Medical treatment methods
 - Inventions relating to atomic energy
- c) **Trademark:** A **distinctive sign, logo, word, name, or symbol** legally registered to identify and distinguish goods or services of a particular source. Once a brand name and mark are registered and legalized it becomes trademark.
- **Legal Reference:** Governed under the **Trademark Act, 1999**
 - Registration is **not mandatory**, but it gives **exclusive legal rights**
 - **Purpose:**

- To protect the **brand identity**
- To prevent misuse or imitation
- Helps consumers identify the origin of a product
- Example: The Apple logo is trademarked.

3. Commercialization

a) Commercializing Research Findings: Process of transforming research discoveries into marketable products. **Steps:**

1. Patent the invention.
2. Partner with industries.
3. Develop prototypes.
4. Launch in the market.

b) Licensing and Royalties

- **Licensing:** Transferring the right to use intellectual property to a third party in exchange for compensation.
- **Royalties:** Payments made to the creator for using their IPR (e.g., book authors receive royalties from publishers).

TRIPS

TRIPS stands for Trade-Related Aspects of Intellectual Property Rights. It is an international legal agreement between all member nations of the World Trade Organization (WTO).

- TRIPS sets minimum standards for various forms of intellectual property (IP) protection, such as copyrights, patents, trademarks, and geographical indications.
 - TRIPS harmonized global IP laws to reduce trade distortion and caused by inconsistent protection standards in different countries.
 - Safeguard creators and other producers of their IP by granting them certain time limited rights to control the use.
- **Copyrights** Protection of literary, artistic, and software works for at least 50 years
 - **Patents** At least 20 years of patent protection for inventions, including in pharmaceuticals
 - **Trademarks** Exclusive rights to use a mark (logo, symbol) and protection against imitation
 - **Geographical Indications (GI)** Protects names/labels linked to regions (e.g., Darjeeling tea)
 - **Industrial Designs** Protection of product designs that are novel and original
 - **Trade Secrets** Protection of confidential business information (e.g., formulas, methods)
 - **Layout Designs of Integrated Circuits** Protection for design of chips and semiconductors

Principles of TRIPS

1. **National Treatment:** Countries must treat foreign IP holders like their own citizens.
2. **Most-Favoured-Nation (MFN):** Any privilege granted to one country must be extended to all WTO members.
3. **Minimum Standards:** Countries must follow minimum standards of IP protection.
4. **Enforcement:** Effective legal mechanisms to ensure protection and resolve IP disputes.

TRIPS and India

- India amended several IP laws (like the **Patent Act**, **Copyright Act**, and **Trademarks Act**) to comply with TRIPS.
- Post-2005, India allowed **product patents** for pharmaceuticals — a major shift.

IMRAD Concept and Design of Research Paper

1. **IMRAD Structure:** IMRAD stands for **I**ntroduction, **M**ethodology, **R**esults, and **D**iscussion—a standard format for structuring research papers.

a) Introduction

- Defines the research problem and objectives.
- Reviews relevant literature.
- Justifies the need for the study, gap.
- Ends with a research question or hypothesis.

b) Methodology

- Describes how the study was conducted.
- Includes details on:
 - Data collection methods.
 - Sampling techniques.
 - Experimental setup (if applicable).
 - Statistical tools used.

c) Results

- Presents the findings **without interpretation**.
- Uses tables, graphs, and statistical data.
- Provides objective reporting of observed trends.
- Assess, explain and evaluate the findings.

d) Discussion

- Interprets the results in context of the research question.
- Compares findings with previous studies.
- Highlights limitations and future research directions.
- Discuss and generalize the result
- State practical implication, strong and weak aspects.
- If more research is needed

Citation

A **citation** is a **formal reference** to a source (book, article, website, etc.) used in your work to **give credit** to the original author and help readers locate the source. **Why Citation is Important:**

- Avoids **plagiarism**
- Supports your arguments with credible evidence
- Shows respect for others' work
- Enhances your credibility as a researcher

□ Common Citation Styles:

Style	Common In
APA (American Psychological Association)	Psychology, Education, Social Sciences
MLA (Modern Language Association)	Literature, Arts, Humanities
Chicago	History, Business, Fine Arts
IEEE	Engineering and Computer Science

▣ Example (APA Style):

Sharma, A. (2020). *Basics of Research Methodology*. Delhi: Academic Press.

Acknowledgement

An **acknowledgement** is a brief note in your thesis, project, or paper that **expresses gratitude** to people or organizations who helped you in your research or project. **Why Include It?**

- Shows professionalism and courtesy
- Gives credit to advisors, guides, funding bodies, peers, etc.
- Acknowledges non-academic support too (like family or friends)

Acknowledgments should recognize:

- Funding agencies.
- Institutions supporting research.
- Individuals who helped (but are not co-authors).

Example: *“I sincerely thank Prof. R. Kumar for his invaluable guidance and constructive feedback. I also appreciate the support of my peers and family during this research.”*

Plagiarism

Plagiarism is using someone else's work, ideas, or words without proper credit, presenting them as one's own.

Types of Plagiarism:

- **Direct Plagiarism** – Copy-pasting without citation.
- **Self-Plagiarism** – Reusing one's previous work without acknowledgment.
- **Paraphrasing Plagiarism** – Rewriting content without citation.
- **Accidental Plagiarism** – Failing to cite sources properly due to negligence.

How to Avoid Plagiarism

- **Proper citation** using APA, MLA, or other styles.
- **Quoting** – Using exact words within quotation marks.
- **Paraphrasing** – Writing in one's own words while crediting the source.
- **Referencing all sources** – Including a bibliography or reference list.
- **Using plagiarism detection tools** before submission.

Reproducibility and Accountability

1. **Ensuring Reproducibility of Research:** Reproducibility means another researcher can replicate the study and obtain similar results. Ways to ensure reproducibility:
 - Providing complete methodology details – Clear steps, sample selection, and tools used.
 - Sharing datasets and code – Open-access repositories like GitHub, Zenodo, and Dryad.
 - Using standardized research protocols – Following established guidelines.
 - Publishing detailed supplementary materials – Extra data, formulas, and algorithms.
2. **Accountability in Publishing:** Researchers must maintain ethical and professional responsibility in publishing.
 - Honest reporting – Avoid falsification or fabrication of data.
 - Proper authorship – All contributors should be fairly credited.
 - Conflict of interest disclosure – Mention any financial or personal influences.
 - Ethical peer review – Fair, unbiased, and constructive review process.